

A dilation obstruction to the displayed vertex-volume bound

Disproof of conjecture 00000007801

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The source defines $G_k(K)$ as the mean volume of the k -simplices spanned by the vertices of a d -dimensional convex body, and explicitly asserts

$$G_k(K) \leq \binom{d}{k} V(S),$$

where S is the standard simplex. Both language versions give this displayed inequality. In dimension $d = k = 1$, an actual interval of length two disproves it. There is no volume-normalization hypothesis on K in the source. This concerns the stated formula, rather than a separately formulated classical Godbersen inequality.

Actual full-dimensional convex body. Use the ordinary real Euclidean line and set

$$K = [0, 2], \quad S = [0, 1].$$

These are compact convex bodies. The interval K has nonempty interior: the point 1 belongs to $(0, 2)$. Its actual ambient real dimension is one. The standard one-simplex S has ordinary Lebesgue volume one.

Exhaustive vertices of K . Its extreme points are exactly 0 and 2. To verify the left endpoint, suppose

$$0 = ax + by, \quad a, b > 0, \quad a + b = 1, \quad x, y \in [0, 2].$$

Both terms are nonnegative, so $ax = 0$ and therefore $x = 0$; likewise $y = 0$. Thus 0 is extreme. For the right endpoint, an analogous representation $2 = ax + by$ gives

$$a(2 - x) + b(2 - y) = 0.$$

Both terms are nonnegative, so $x = y = 2$.

Conversely, if $0 < z < 2$, then

$$z = (1 - z/2) 0 + (z/2) 2$$

is a strict convex combination of distinct endpoints. Hence z is not extreme. The complete vertex set is therefore $\{0, 2\}$.

Complete mean of actual vertex simplices. A one-simplex requires two distinct vertices. There is exactly one unordered two-element subset of $\{0, 2\}$, namely the whole vertex set. The corresponding simplex is its genuine convex hull:

$$\text{conv}\{0, 2\} = [0, 2] = K.$$

Therefore the mean is a sum over one actual simplex divided by the number of such simplices:

$$G_1(K) = \frac{V(\text{conv}\{0, 2\})}{1} = V([0, 2]) = 2.$$

Using ordered distinct vertex pairs would give two identical volumes and the same mean, so no averaging convention affects this example.

Failure of the displayed inequality. The source's right-hand side is

$$\binom{1}{1} V(S) = 1 \cdot V([0, 1]) = 1.$$

Thus its claimed inequality is the false statement

$$2 \leq 1.$$

The source also displays the proposed coefficient

$$c_{d,k} = \frac{d!}{(d-k)!k!} \frac{V(S)}{V(K)}.$$

Here this coefficient equals $1/2$, so the original form $G_1(K) \leq c_{1,1} V(K)$ again gives $2 \leq 1$.

Scope and scaling. Dilation makes the defect transparent: for $K = [0, L]$ the sole vertex one-simplex has volume L , while the fixed standard simplex continues to have volume one. The actual counterexample uses $L = 2$ and satisfies all stated convex-body conditions. The special middle-dimensional stability assertion is a separate conjunct and is not needed. No disproof of a different normalized or mixed-volume formulation is claimed.

Formal correspondence and reproduction. Lean uses the actual real interval, compactness, convexity, interior and real dimension. It proves the complete set of Mathlib extreme points, a bijective finite enumeration of the vertices, and exhaustive enumeration of every two-vertex subset. Each simplex is the actual convex hull of its selected vertices. Its volume is actual Lebesgue measure converted to a real number. The finite mean, both actual interval volumes, the displayed coefficient and the failure of both displayed inequalities are proved.

Run `lake build` in `lean/` with Lean 4.19.0 and public Mathlib revision:

`c44e0c8ee63ca166450922a373c7409c5d26b00b`

The project prints the relevant axiom audits.